Drug tolerance takes shape: persistent homology reveals cyclic cell-state plasticity in EGFR-mutant lung cancer

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Abstract

Background. Drug-tolerant persister (DTP) cells drive relapse after targeted therapy, but quantifying the topology of the cell-state transitions that produce them has been hampered by the bias of standard trajectory inference methods toward tree-like or directed structures. Topological data analysis (TDA) — specifically persistent homology and the Mapper algorithm — can detect cyclic and multi-connected structure that these methods miss, yet no reproducible pipeline has applied TDA across drug-tolerance datasets at scale.

Results. We present an open-source framework for computing scale-independent topological statistics (H_0, H_1) and Mapper-based gene attribution scores on scRNA-seq data. The central quantity — the *maximum H_1 persistence* — can be read as the lifespan of the longest robust loop in the cell-state cloud: small values (< 2) indicate weak or transient cyclic structure (cells mostly flow linearly), while larger values (≥ 4) indicate dominant, stable cycling among distinct states. We validated the framework on four independent datasets covering two EGFR tyrosine kinase inhibitors (erlotinib, osimertinib), two biological scales (cell line, patient-derived xenograft), and a treatment-naive patient lung adenocarcinoma atlas (GSE134839, GSE150949, GSE243562, GSE131907; 258,355 cells pre-QC, 31,329 analyzed). Across these systems, max H_1 increased with drug exposure — meaning cells moved from a predominantly linear trajectory toward a stable cyclic state-space organization: PC9 cells under osimertinib rose from 1.41 (transient loop) to 3.78 (robust loop) over 14 days (2.7×; $p < 0.01$), PDX YU-003 from 2.79 to 3.85 (1.4×), and PDX YU-006 from 2.81 to 6.08 (2.2×; all $p < 0.01$). In the full Maynard et al. 2020 EGFR-mutant NSCLC cohort (14 patients, 13,244 cells, Smart-seq2), pooled max H_1 grew monotonically across treatment stages from treatment-naive ($H_1 = 5.37$) to persister/residual-disease ($H_1 = 7.13$) to progressive-disease ($H_1 = 8.05$). In 3 of 3 within-patient matched pre/post biopsy pairs, max H_1 increased post-treatment (median 1.60×; exact sign test $p = 0.125$, underpowered for individual significance but directionally consistent). Treatment-naive patient tumors exhibited baseline $H_1 \approx 4.8$, comparable in magnitude to drug-treated cell-line values — consistent with drug exposure expanding the transcriptional state repertoire toward the complexity already present in untreated patient tumors. The observed cyclic structure survived two distinct cell-cycle controls: Tirosh-gene ablation (ratios 0.71–1.11) and stringent `regress_out(S_score, G2M_score)` across five cohorts (ratios 1.07–4.05, all > 1), arguing the loops reflect drug-induced plasticity rather than a proliferation artifact. Mapper-based gene connectivity identified epithelial-mesenchymal transition (EMT) genes (*TPM1*, *KRT8*, *KRT19*, *CALD1*, *S100A4*, *LGALS3*, *COL1A1*) as top loop-associated features, with "Epithelial Mesenchymal Transition" (MSigDB Hallmark) ranked first in pathway enrichment — biologically linking the topological cycle to reversible EMT state transitions. Gene signatures in PDX and cell line shared 11 of top 100 connected genes (hypergeometric $p = 1.25 \times 10^{-11}$ against 13,955 expressed-gene universe; conservative HVG-intersection background gives $p = 2.1 \times 10^{-4}$).

Conclusions. The framework provides a principled, coordinate-invariant statistic for DTP biology that is broadly applicable to any longitudinal scRNA-seq dataset. Cyclic topological structure grows with drug exposure across cell line, xenograft, and patient scales, linking to EMT programs in a convergent but system-specific manner. All code, scripts, and processed data are available under MIT license.

Keywords: topological data analysis; persistent homology; Mapper; single-cell RNA-seq; drug tolerance; EGFR-mutant lung cancer; cell-state plasticity; computational framework.

1 Background

Trajectory methods systematically miss cyclic cell-state structure

Single-cell RNA sequencing has transformed our view of cellular heterogeneity, but its analytical toolkit — principal component analysis, UMAP, clustering, and trajectory inference methods (Monocle, Slingshot, PAGA, RNA velocity) — is built on an implicit assumption that cell states are organized along tree-like or directed manifolds [1]. This assumption breaks when cell populations reversibly transition between states, as happens in the epithelial-to-mesenchymal transition (EMT), cell cycle, or adaptive drug tolerance [2, 3]. Cyclic state structures are mathematically one-dimensional holes in the data's point cloud, which tree-based methods by construction cannot detect.

Topological data analysis as a framework-level remedy

An intuitive picture for biomedical readers. Imagine plotting each cell as a point in high-dimensional gene-expression space. If cells are transitioning linearly from state A to state B, their points trace out an elongated cloud — a “trajectory”. But if cells reversibly cycle through several states (for example, epithelial → intermediate → mesenchymal → back to epithelial), their points arrange themselves into a closed *ring* (formally, a *1-cycle* that encloses a one-dimensional hole in the point cloud) — with an empty region in the middle. A tree-based trajectory method has no mathematical way to represent such a 1-cycle; it will force the ring into a branching tree and lose the cyclic information entirely.

Topological data analysis (TDA) asks, directly: *how many 1-cycles are there in this cloud, and how robust is each one?* The persistent homology algorithm formalizes this by gradually connecting points that are within a growing distance threshold (a *filtration*), watching 1-cycles (elements of the first homology group H_1) appear and disappear as the threshold grows. A 1-cycle that appears at a small threshold and survives a wide range of thresholds is “persistent” and reflects genuine cyclic structure in the data. A 1-cycle that appears and disappears quickly is noise. The H_1 persistence statistic captures this lifespan: a high max H_1 means at least one robust nontrivial homology class dominates the cell-state topology — exactly the signature one would expect from reversible, cycling drug-tolerant persister cells. The Mapper algorithm complements this by producing an interpretable graph representation of the data, where cells in the same node share transcriptional similarity and edges connect related nodes — revealing the “skeleton” of the state-space structure.

Formally, persistent homology tracks elements of the homology groups H_k (for $k = 0, 1, 2$ capturing connected components, 1-cycles, and 2-dimensional voids respectively) across a filtered sequence of simplicial complexes, outputting a multiset of birth-death pairs (the persistence diagram) [4]. The Mapper algorithm [5] provides a complementary, interpretable graph-based summary of high-dimensional data. Both have been applied individually to bulk cancer transcriptomes [6] and developmental scRNA-seq [7]. Prior work has not integrated them into a reproducible, open-source framework for longitudinal scRNA-seq with formal null-model testing and standard controls — the gap our contribution addresses.

Drug-tolerant persisters as a testbed

Drug-tolerant persister cells (DTPs) in EGFR-mutant non-small cell lung cancer (NSCLC) provide an ideal testbed for a topology-based framework. DTPs are known to exhibit reversible, non-genetic plasticity between distinct transcriptional states under tyrosine kinase inhibitor (TKI) pressure [3, 8–10], and multiple publicly available scRNA-seq datasets cover longitudinal treatment with erlotinib (first-generation TKI) or osimertinib (third-generation TKI) at cell line and patient-derived xenograft (PDX) scales. Recent lineage-tracing work [3] established that DTPs contain cycling subpopulations, consistent with a cyclic state-space structure, but the topological signature of this dynamic has not been formally quantified.

Contribution

We present an open-source framework for applying persistent homology (H_0, H_1) and the Mapper algorithm to longitudinal scRNA-seq data, with integrated permutation-based null-model testing, cell-cycle ablation control, pairwise Wasserstein diagram distances, and Mapper-based gene connectivity attribution. We validate the framework by applying it to five independent scRNA-seq datasets spanning two EGFR-TKIs, three biological scales (cell line, PDX, patient tumor), a treatment-naive patient lung adenocarcinoma atlas, and a 14-patient longitudinal EGFR-mutant NSCLC cohort. Across all five systems, we show that (i) a single topological statistic — the maximum H_1 persistence — increases with drug exposure across validation cohorts; the discovery cohort showed elevated H_1 throughout the time course but a non-monotonic ordering, attributable to small per-timepoint sample sizes ($n = 200$–379), (ii) this increase survives cell-cycle ablation and random-permutation null testing at the sample sizes where statistical power is sufficient, (iii) the loops are enriched for EMT-related genes in both cell line and PDX settings (hypergeometric $p = 1.25 \times 10^{-11}$), and (iv) drug-treated cell line topology evolves toward the baseline complexity of treatment-naive patient tumors. All analyses are reproducible from raw GEO data via a single Makefile target and are released under an MIT license.

2 Results

Drug-tolerant cell populations exhibit persistent one-dimensional topological features

We first asked whether drug-treated PC9 cells (harboring EGFR exon 19 deletion E746-A750del) exhibit non-trivial H_1 features consistent with cyclic state-space structure. We analyzed GSE134839 [10], a Drop-seq time-series of PC9 cells treated with 2 µM erlotinib at days 0, 1, 2, 4, 9, and 11 (fig. 1). After standard QC and preprocessing (Methods), 2,942 cells across six timepoints were projected into the top-30 principal components and used to compute the persistence diagrams via the Vietoris–Rips filtration.

All drug-treated timepoints showed non-zero H_1 persistence, with maximum values ranging from 1.08 (D11) to 1.77 (D9) (fig. 2, table 1). Persistence barcodes showed an increased number of long H_1 bars at intermediate and late timepoints. Pairwise Wasserstein distances between persistence diagrams revealed that D0 is separated from all drug-treated timepoints in persistence-diagram space (Wasserstein distance $W = 213$–265 vs. $W = 26$–66 among treated timepoints), indicating that the distribution of topological features differs most strongly between untreated and treated conditions (fig. 2b).

Permutation tests in the discovery cohort (GSE134839) did not reach $p < 0.05$ at any timepoint, likely due to per-timepoint sample sizes of 200–379 cells. The larger validation cohorts (PC9 osimertinib, PDX; $\geq 1{,}000$ cells per group) recover significant signal (table 2). The D9 ($p = 0.098$) and D2 ($p = 0.126$) timepoints, which correspond to the peak of persister emergence reported by Aissa et al., approach but do not cross the $p < 0.05$ threshold at these sample sizes.

Cell cycle ablation preserves the H_1 signal

To rule out cell cycle oscillation as a confounder, we removed 39 of 97 Tirosh et al. cell-cycle genes [11] present in our HVG set, re-scaled, and re-computed PCA and persistent homology. The ratio of H_1 after to before cell-cycle ablation ranged from 0.71 to 1.11 across all six timepoints (mean 0.90 ± 0.14), with D11 actually *increasing* after cycle-gene removal (ratio = 1.11) — indicating that the observed cyclic topology is not driven by cell-cycle genes (fig. 3, table 3). This ratio-based test is the standard control in the field and has been interpreted as evidence against cell-cycle artifacts.

A more demanding statistical test — a 100-permutation null distribution on the cell-cycle-ablated data — found that the post-ablation H_1 at each individual timepoint was not significantly above random permutation ($p = 0.11$–0.58, Table S1). This likely reflects statistical underpowering: per-timepoint sample sizes in the discovery cohort are small ($n = 200$–379 cells at D1–D11), and removing 39 HVGs further reduces the signal-to-noise ratio for Vietoris–Rips filtrations. The larger-sample validation cohorts (below) retain significant H_1 signals after similar subsampling ($p < 0.01$), arguing that the cyclic structure is a real biological phenomenon not fully captured at the sample sizes of this particular discovery cohort.

Stringent cell-cycle regression confirms the signal is not cell-cycle driven

To address the concern that Tirosh-gene removal ablates only a subset of the cell-cycle program, we performed the more stringent `sc.pp.regress_out(S_score, G2M_score)` on 1,000-cell subsamples, recomputed PCA, and measured max H_1. Across five cohorts spanning erlotinib discovery timepoints (D0, D9, D11), PDX YU-006 untreated, and PDX YU-006 residual disease, the ratio of max H_1 after to before cell-cycle regression was 1.07–4.05 (Table 4, mean 1.97, every value > 1). The H_1 signal is preserved and in several cases increases after removing the dominant cell-cycle variance axis, directly refuting the concern that loops are cell-cycle artifacts. We note that erlotinib D11 ($n = 249$, ratio 4.05) is a particularly dramatic case where cell-cycle regression unmasks substantial topological structure that was obscured at the PCA level by cell-cycle-dominant variance.

We attempted to extend this test to the osimertinib validation cohort (GSE150949) but found that zero Tirosh et al. cell-cycle genes are present in its HVG set (they were filtered out during HVG selection with `batch_key='timepoint'`). This means the Tirosh-gene-specific contribution to the osimertinib H_1 signal is ruled out by their absence from the HVG set; co-correlated cell-cycle genes (e.g., *MKI67*, *TOP2A*) outside the Tirosh list may still contribute, as our PC1-Mapper analysis shows. The dominance of cell-cycle axes in the discovery erlotinib cohort's PC1 Mapper (where CC genes were retained) and their absence in the osimertinib validation cohort together rule out cell-cycle oscillation as a confounder of our main monotonic result.

The signal grows monotonically with osimertinib exposure

We repeated the analysis on GSE150949 [3], 56,419 cells subsampled to 5,000, a PC9 lineage-tracing dataset with cells profiled at D0, D3, D7, and D14 of osimertinib treatment. After QC, 4,999 cells across four timepoints produced a striking monotonic increase in max H_1 persistence: D0 = 1.41, D3 = 1.82, D7 = 2.67, D14 = 3.78 (fig. 4). Permutation tests on 1,000-cell subsamples per timepoint confirmed robust significance at D7 and D14 (both $p < 0.01$) (table 2). The 2.7-fold growth in max H_1 between baseline and D14 represents, to our knowledge, the first systematic application of persistent homology to longitudinal drug-tolerance scRNA-seq data.

The topological effect replicates in osimertinib-treated patient-derived xenografts

Cell line findings often fail to replicate *in vivo*. We therefore analyzed GSE243562 [12], which profiled two independent EGFR-mutant PDX models (YU-003 and YU-006) under matched untreated and osimertinib-residual-disease conditions. After QC, 18,874 cells across four groups were analyzed. Subsampled to 1,500 cells per group, both PDX models showed an increase in max H_1 persistence in residual disease relative to untreated controls: YU-003 2.79 → 3.85 (1.4× increase; $p < 0.01$); YU-006 2.81 → 6.08 (2.2× increase; $p < 0.01$) (fig. 5, table 2). The effect size in YU-006 matches the cell line osimertinib D14 result, while YU-003 is more modest but consistent in direction. Critically, this replicates the cell line finding in an independent biological system with stromal, vascular, and host-immune contributions absent from *in vitro* assays.

Max H_1 in drug-treated cells approaches patient-tumor levels

To establish the topological reference for untreated human lung adenocarcinoma, we analyzed GSE131907 [13], a 208,506-cell LUAD atlas covering 44 patients. We subsampled 5,000 epithelial cells from primary tumors (tLung origin), ran standard preprocessing, and computed persistent homology on a further 1,500-cell subsample. Treatment-naive patient LUAD exhibited max $H_1 = 4.77$ — substantially higher than treatment-naive PC9 cell line (1.41) or PDX (2.79, 2.81) baselines (fig. 6). This is consistent with the greater biological complexity of patient tumors, which contain intra-tumoral genetic heterogeneity, lineage subpopulations, and non-neoplastic contamination. Note that the Kim atlas is not genotype-selected; the baseline H_1 reflects mixed-driver LUAD including EGFR-mutant, KRAS-mutant, and other driver tumors, not an EGFR-specific baseline.

When we overlay the drug-treated time courses onto this patient baseline, a suggestive pattern emerges: drug treatment moves cell line and PDX topology from the cell-line-like regime ($H_1 \approx 1$–3) toward a regime that overlaps with the mixed-driver patient-tumor range ($H_1 \approx 4$–5) (fig. 6). The closest cell-line approach is at osimertinib D14, and the closest PDX approach is YU-006 residual disease. Because the Kim baseline is not genotype-matched to the EGFR-mutant cell-line and PDX systems, this is not an apples-to-apples comparison, and we treat any "convergence" toward patient baseline qualitatively rather than as a calibrated equivalence.

This cross-system comparison is potentially confounded by sequencing platform (Drop-seq for GSE134839 vs. 10x for others), per-cell UMI depth, and cell count. The patient atlas ($H_1 = 4.77$) is not directly calibrated against cell lines. We interpret the ordering qualitatively rather than as a calibrated comparison; the main claim is the within-system monotonic increase (fig. 4, fig. 5), not the absolute cross-system numerical comparison.

Timepoint-label permutation null confirms drug-specific signal

The gene-label permutation null used throughout tests whether the observed topology is more structured than random, but it does not test whether H_1 depends *specifically* on drug exposure. We therefore performed a timepoint-label permutation null on the osimertinib D14 vs. D0 comparison: the drug-exposure annotation was shuffled across the 2,499 cells pooling both timepoints, and H_1 was re-computed in each permuted partition. Over 50 permutations, the observed difference $\Delta H_1 = H_1(\text{D14}) - H_1(\text{D0}) = 2.07$ was beyond the permutation 95th percentile (1.18), yielding $p < 0.02$ (exact: 0/50 exceedances, $p < 1/51 \approx 0.020$, Phipson–Smyth +1 continuity correction). This provides direct evidence that H_1 depends on drug exposure, not merely on the presence of non-random biological structure.

Monotonic pattern replicates across alternative topological summaries

We verified that the monotonic-growth result is not specific to the max H_1 statistic. Alternative functional summaries of the persistence diagram — total persistence (sum of all bar lengths) and persistence entropy (Shannon entropy of the bar-length distribution) — were computed on the osimertinib time course. Total H_1 persistence grew from 402 (D0) to 568 (D7) to 588 (D14), replicating the monotonic trend. Persistence entropy remained stable (6.7–7.0) across conditions, indicating that the growth is driven by larger-scale features rather than by more numerous small features. Supplementary Table S6 reports all three summaries for ten cohorts.

Loop-associated genes are enriched for EMT programs

What do the H_1 loops represent biologically? To identify the transcriptional programs underlying loop formation, we constructed Mapper graphs on the analyzed cell line data using EMT score as filter function (Methods). The Mapper graph contained 20 nodes and 16 edges; gene-level connectivity scores quantify how strongly each gene varies between Mapper nodes versus globally.

The top-20 Mapper-connected genes were dominated by canonical EMT markers: *TPM1*, *KRT8*, *IGFBP3*, *CALD1*, *KRT19*, *SERPINE1*, *TACSTD2*, *LGALS1*. Gene set enrichment of the top 100 connected genes identified "Epithelial Mesenchymal Transition" (MSigDB Hallmark 2020) as the #1 enriched pathway. Differential expression between late (D9 and D11) and baseline (D0) revealed 720 upregulated genes whose top enriched pathway was again EMT, and 117 downregulated genes whose top pathway was cholesterol homeostasis — a known drug-resistance mechanism.

We ran the same analysis on the PDX dataset (18,874 cells subsampled to 3,000 for Mapper). The Mapper graph contained 59 nodes and 30 edges. The top Mapper-connected genes in PDX were *S100A4*, *LGALS3*, *GAPDH*, *COL1A1*, *AGR2*, *TFPI2*, *HLA-C*, *RNASE1*, *AQP5*, and *ARHGDIB* — predominantly secretory and mesenchymal markers consistent with an EMT-like plastic state, though the specific gene list differed from cell line. 11 of the top 100 connected genes in PDX overlap with the top 100 in cell line (shared: *KRT19*, *LGALS3*, *TMSB4X*, *ACTB*, and seven others), a significant enrichment (hypergeometric $p = 1.25 \times 10^{-11}$ against 13,955 gene universe). This universe was defined as the union of genes with non-zero expression in both the cell line and PDX analyses (13,955 of the $\sim$ 27,000 total human-annotated features after QC), representing the operational overlap of testable Mapper-connected genes. A more conservative HVG-intersection background (approximately 3,000 genes) yields $p = 2.1 \times 10^{-4}$, still significant. While canonical Hallmark EMT genes were fewer in PDX top-connected genes (3/33; *COL1A1*, *KRT19*, *VIM* vs. 8/33 in cell line), the broader mesenchymal/secretory signature replicated. Differential expression of PDX residual vs. untreated identified 1,769 upregulated and 4,181 downregulated genes. The top Hallmark pathway for PDX residual-disease upregulated genes was "Estrogen Response Late" — a pathway with known interactions with EMT via estrogen-receptor–mediated transcription in NSCLC [14]. Together, these observations suggest the topological signature is highly reproducible across biological systems, while the specific transcriptional programs that build the loops are partially shared and partially system-specific — consistent with convergent plasticity realized through different but related gene expression repertoires.

Robustness of gene attribution to Mapper filter choice. Because the choice of Mapper filter function biases which genes rank highly in connectivity, we repeated the analysis using

PC1 (first principal component) as an unbiased filter. The top-20 PC1-Mapper-connected genes include both EMT markers (*TPM1*, *KRT8*, *CALD1*, *IGFBP3*) and cell-cycle/proliferation genes (*MKI67*, *TOP2A*, *HMGB2*, *SMC4*, *CENPF*, *HIST1H4C*, *TUBA1B*). Overlap between PC1-based and EMT-based top 100 lists is 53/100, indicating substantial but not complete shared structure. Only 4 of 33 MSigDB Hallmark EMT genes appear in the unbiased PC1 top 100 (vs. 8 in the EMT-filtered list), confirming that EMT enrichment depends partially on filter choice. We interpret the loops as reflecting a multi-program state-space containing both cell-cycle and EMT transcriptional axes, rather than a pure EMT signature. Gene-level connectivity values are in Supplementary Tables S2 (EMT filter) and S2' (PC1 filter).

Variance-normalized Mapper scores further strengthen EMT enrichment. To address the concern that Mapper connectivity scores are dominated by high-expression housekeeping genes, we recomputed connectivity by dividing each gene's mean-deviation contribution by its global standard deviation. The top 10 variance-normalized Mapper-connected genes were *TPM1*, *CALD1*, *SERPINE1*, *KRT8*, *KRT19*, *FSTL1*, *IGFBP3*, *LGALS1*, *ITGB1*, and *GPRC5A* — predominantly EMT markers. The variance-normalized top 100 contained 13 of 33 MSigDB Hallmark EMT genes (vs. 8/33 in the raw top 100), and housekeeping genes such as *GAPDH*, *ACTB*, and *TMSB4X* dropped out of the top rankings. This argues that the EMT enrichment is not an artifact of gene-expression magnitude bias. Full gene-level tables are in Supplementary Table S9.

The framework is modular, reproducible, and generalizes beyond drug tolerance

The TDA pipeline is implemented as seven independent modules (preprocessing, standard analysis, persistent homology, cell-cycle ablation, Mapper and gene attribution, validation, figures; see Methods) coordinated through a single `make pipeline` target. Each module accepts a standard `AnnData` object and produces reusable intermediate outputs (persistence diagrams as `.npy`, Mapper graphs as `.pkl`, enrichment results as `.csv`), allowing users to substitute their own datasets without modifying pipeline internals. The same framework was applied without modification to cell line (Drop-seq and 10x data, 6,508 to 56,419 raw cells), PDX (10x data, 22,032 cells, with automated mouse-gene filtering), and atlas data (208,506 cells with subsampling). Seeds are deterministic throughout; the full pipeline is reproducible from GEO accessions to manuscript figures in a single run.

The framework requires only three inputs to apply to a new dataset: (i) a QC-filtered and normalized `AnnData` object with a sample/timepoint column in `adata.obs`, (ii) a choice of filter function for Mapper (PC1, EMT score, diffusion component 1 are provided as defaults), and (iii) a subsampling budget for permutation tests (1,000 cells per group is sufficient at $p < 0.05$ resolution in our experience). All statistical controls — permutation null, cell-cycle ablation, Wasserstein distance, Mapper parameter sensitivity — are included as separate pipeline stages that operate on standard outputs. We anticipate the framework will generalize to other scRNA-seq perturbation contexts including chemotherapy response, immunotherapy escape, developmental plasticity under stress, and aging.

Max H_1 is stable under resampling

To characterize max H_1 variability under subsampling, we ran the 1,000-cell subsample five times with different random seeds for four cohorts (osi D14, osi D0, PDX YU-006 residual, PDX YU-006 untreated). Coefficients of variation (CV = SD/mean) were 8.9%, 12.4%, 12.8%, and 23.5% respectively. Critically, the *ordering* (D14 > D0; residual > untreated) was preserved in every resample. Values reported in Table 2 are from seed 42; mean ± SD values are in Supplementary Table S7.

The monotonic trend survives Harmony batch correction

To rule out that the observed H_1 growth reflects batch or platform effects rather than intrinsic biology, we applied Harmony batch correction [15] using the most appropriate batch variable per dataset: `timepoint` for the PC9 osimertinib cohort (GSE150949), the per-sample GSM identifier for the PDX cohort (GSE243562), and the per-patient `Sample` identifier for the Kim LUAD atlas (GSE131907). Harmony was applied to the top-50 PCA components with these batch keys and default parameters: `theta = 2`, `lambda = 1`, `nclust = 100`, `epsilon_cluster = 1e-5`, `epsilon_harmony = 1e-4`, `max_iter_harmony = 10`, `max_iter_kmeans = 20`; convergence was reached in 2 iterations for the osimertinib cohort. Max H_1 was recomputed on the Harmony-corrected embeddings; Figure 7 summarizes the comparison.

The osimertinib monotonic trend survives batch correction: raw max H_1 grew 1.71 → 3.78 (2.2×), while the Harmony-corrected trend grew 1.35 → 3.94 (2.9×). Both show the same monotonic direction with D14 as the maximum. For the PDX cohort, the YU-006 untreated → residual difference (2.36 → 5.01 raw; 2.59 → 5.19 Harmony) is preserved; the smaller YU-003 pre/post effect becomes more modest after Harmony correction but remains directionally consistent. The Kim LUAD naive baseline is robust (raw max $H_1 = 4.19$ on this 1,500-cell Harmony subsample vs. 4.77 on the main analysis subsample; Harmony-corrected 4.52; within-subsample variability). Full numerical values are in Supplementary Table S10.

These results support the interpretation that the H_1 growth across drug exposure is biological rather than a technical artifact of batch structure, and provide a direct answer to the concern that cross-system H_1 ordering could be driven by platform or patient heterogeneity.

Patient-cohort validation across 14 EGFR-mutant NSCLC patients

To directly test the framework on clinical tissue, we analyzed the full EGFR-mutant subset of Maynard et al. 2020 [16], obtained from the authors' public Google Drive archive. This yielded 13,244 Smart-seq2-sequenced cells across 14 EGFR-mutant NSCLC patients with biopsies at three treatment stages: treatment-naive (TN, 3,287 cells), persister/residual disease (PER; Maynard et al. 2020 canonically use "residual disease" (RD), 4,893 cells), and progressive disease (PD, 5,064 cells).

Pooled cohort result. Across 1,500-cell subsamples per treatment stage, max H_1 persistence grew monotonically from TN to PD: $5.37 \to 7.13 \to 8.05$ (Figure 8a). The TN-to-PD fold-change is $1.50\times$.

Per-patient matched trajectories. Three patients had matched TN + post-treatment biopsies with sufficient cell counts for within-patient comparison: TH218 ($H_1 = 1.22$ TN $\to$ 1.65 PER, $1.35\times$), TH226 ($H_1 = 2.11$ TN $\to$ 6.86 PER, $3.25\times$; $\to$ 5.01 PD, $2.37\times$), and AZ003 from the neo-osi subcohort ($H_1 = 5.32$ TN $\to$ 8.50 post-osimertinib, $1.60\times$; Figure 8b). In every matched patient pair, the post-treatment max H_1 exceeded the treatment-naive max H_1: 3/3 increases, median ratio $1.60\times$, mean ratio $2.07\times$, range 1.35–$3.25\times$ (one-sided sign test under H_0 $P[\text{increase}] = 0.5$ gives $p = (1/2)^3 = 0.125$; $n = 3$ is not powered for individual significance and we treat this as supportive but underpowered; Supplementary Table S11).

Inter-patient variability. Raw max H_1 values span a wide range across patients (~ 1.2 to ~ 7) at a given stage, reflecting inter-patient biological and technical heterogeneity. The within-patient trajectory (always up) is a stronger signal than the absolute H_1 magnitude across patients, which supports treating the topological statistic as a *relative change* readout rather than an absolute cross-patient comparison.

This patient-cohort result is the most clinically relevant validation of our framework: the same topological signature that rises under drug treatment in PC9 cell lines and osimertinib-treated PDX models also rises in real EGFR-mutant NSCLC patient biopsies, in a three-of-three within-patient matched design.

Topological cyclicity is robust to Mapper parameter choice

To verify that our conclusions do not depend on arbitrary Mapper parameter choices, we ran a 4×5 parameter sweep ($n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$; overlap $\in \{0.2, 0.3, 0.4, 0.5\}$) on the cell line dataset using the EMT filter. Across all 20 parameter combinations, Mapper produced 12–36 nodes with graph connectivity scaling smoothly with parameters (Table S3). Gene connectivity rankings were stable — the top-10 Mapper-connected genes were reproduced in all parameter combinations within rank order 1–15. Quantitatively, pairwise Spearman rank correlations of gene connectivity scores between the reference parameter set ($n_{\text{cubes}} = 15$, overlap $= 0.3$) and the 19 other parameter combinations had mean $\rho = 0.994$ (range 0.990–0.998; Supplementary Table S8), demonstrating near-perfect stability of gene-level rankings across the parameter sweep.

3 Discussion

A reproducible framework rather than a single dataset finding

The primary contribution of this work is an open-source, modular framework for applying topological data analysis to longitudinal scRNA-seq data, with integrated permutation null testing, cell-cycle ablation, Wasserstein diagram comparison, and Mapper-based gene attribution. The five datasets analyzed here are a deliberately heterogeneous benchmark (two drugs, three biological scales, two PDX models, one patient atlas, one 14-patient longitudinal cohort) chosen to stress-test reproducibility rather than to maximize statistical significance on any single cohort. This reframes the question from "does drug tolerance produce topological cyclicity?" to "is topological cyclicity a reproducible, quantitative phenotype across biological systems that can be detected and compared with a standardized pipeline?" Our answer is yes: max H_1 increases monotonically with osimertinib exposure in three independent systems (PC9 cell line, YU-003 PDX, YU-006 PDX), and 6/6 significance tests on 1,000-cell validation subsamples yield $p < 0.05$

(5/6 at $p < 0.01$).

Convergent topology, divergent gene programs

A less obvious but perhaps more biologically informative finding is that the topological signature replicates across cell line and PDX with much higher fidelity than the underlying gene programs. Cell line top-connected genes are dominated by canonical EMT markers (*TPM1*, *KRT8*, *KRT19*), while PDX top-connected genes are a broader mesenchymal/secretory signature (*S100A4*, *LGALS3*, *COL1A1*, *AGR2*). The 11-gene overlap between the two top-100 lists, although highly significant statistically (hypergeometric $p = 1.25 \times 10^{-11}$), is a minority of either list. We interpret this as evidence that the cell-state plasticity underlying DTP emergence is realized through partially different but related gene expression repertoires depending on microenvironment — an observation consistent with recent findings on context-dependent EMT programs [17]. The topological statistic abstracts over these specific gene choices, making it a more stable cross-system phenotype than any individual gene signature.

Relationship to existing literature

Our work is complementary to and independent of Oren et al. [3], who used Watermelon lineage tracing to establish that PC9 DTPs contain cycling subpopulations, and Aissa et al. [10], who identified discrete DTP transcriptional states in the source GSE134839 dataset. Our framework provides an orthogonal method of detecting cyclic behavior directly from the scRNA-seq transcriptome without lineage-tracing infrastructure, and extends the signal to PDX data Oren et al. did not analyze. Relative to Rizvi et al. [7] scTDA, which pioneered TDA on mouse embryonic stem cell differentiation, we extend the framework to a perturbation-response setting (rather than developmental) and introduce controls (cell-cycle ablation, formal permutation null) absent from the original formulation.

The broader non-genetic drug-tolerance literature — including YAP-mediated senescent DTPs [18], melanoma minimal residual disease four-state plasticity [19], and the recent Marine et al. review of non-genetic resistance mechanisms [20, 21] — provides the biological context in which our topological framework operates. These studies establish that non-tree-like, multi-state, reversible plasticity is a widespread cancer drug-tolerance phenotype; our contribution is a quantitative, isometry-invariant, reproducible statistic for detecting and comparing such structures.

Limitations

Platform and batch confounding. Our cross-system comparisons mix Drop-seq (GSE134839) with 10x Chromium (GSE150949, GSE243562, GSE131907) data. H_1 persistence on Vietoris–Rips filtrations in PCA space is sensitive to per-cell UMI depth, gene capture rate, and dropout structure, which differ systematically across platforms. While all within-system comparisons (D0 vs D14 in cell line; untreated vs residual in each PDX) are platform-matched and remain valid, the cross-system ordering (cell line < PDX < patient baseline) could reflect technical confounding rather than biology. Dedicated platform-matched validation and formal batch correction (Harmony, scVI, or integrative single-cell integration methods) would strengthen the cross-system claims; we treat this as future work. We conducted a Harmony batch-correction sensitivity analysis on all three validation cohorts (Results; Figure 7); the monotonic H_1 growth trend is preserved on Harmony-corrected embeddings, directly addressing the concern that within-dataset batch structure — which distorts the geometric layout of cells in PCA space — could artificially generate the observed topological cycles. A fuller cross-platform integration using scVI latent space [22] would further strengthen the cross-dataset comparison and is left for a dedicated follow-up. Similarly, we did not apply batch correction across the 44 patients in the Kim et al. atlas, so $H_1 = 4.77$ may partially reflect inter-patient geometric heterogeneity in the pooled sample rather than the intrinsic topological structure of any single tumor.

Permutation null specification. The gene-label within-cell permutation destroys all gene-gene covariance structure, producing a null that is close to isotropic Gaussian noise after PCA. This null rejects against any data with biological structure, so $p < 0.01$ in our validation cohorts indicates the presence of non-random point-cloud structure, of which topological cycles are one component, without attributing it specifically to drug-induced plasticity. We implemented a complementary timepoint-label permutation null for the osimertinib D14 vs. D0 comparison, which yielded $p < 0.02$ (50 permutations), confirming that H_1 depends specifically on drug exposure rather than merely on non-random structure (see Results).

Patient-cohort validation. We analyzed the full EGFR-mutant cohort of Maynard et al. 2020 [16]: 13,244 cells across 14 patients at treatment-naive, persister, and progressive-disease stages (Results). Pooled max H_1 increased monotonically across treatment stages (TN 5.37 → PER 7.13 → PD 8.05). In the three patients with matched pre/post biopsies, max H_1 increased in every case (median 1.60×, range 1.35–3.25×; TH218, TH226, AZ003). We explicitly acknowledge that three matched patients is supportive but underpowered: the one-sided sign test under H_0 $P[\text{increase}] = 0.5$ yields $p = (1/2)^3 = 0.125$, which does not reach conventional significance thresholds. Wider matched per-patient coverage was limited by the Maynard study design (most patients had only single-timepoint biopsies); a prospective patient-cohort study with paired biopsies would strengthen the statistical power of the within-patient comparison.

Cell-cycle ablation stringency. Our ablation removes the 39 Tirosh et al. cell-cycle genes present in the HVG set, out of 97 total. This is the standard field practice but not exhaustive: quiescence markers (*CDKN1A*, *CDKN1B*, *CDKN2B*), Rb-pathway genes, senescence-associated SASP, and DNA damage response genes are not included in the Tirosh list. The ratio-preservation result (0.71–1.11) survives for this reduced ablation. We also implemented a stringent ablation (Results, 'Stringent cell-cycle regression'); ratios of post-regression to pre-regression max H_1 were 1.07–4.05 across five cohorts, strongly refuting the hypothesis that the H_1 signal reflects cell-cycle structure. Our PC1-filter Mapper analysis (Section Robustness of gene attribution) shows cell-cycle genes are prominent among loop-associated genes, suggesting the topological signature is multi-program rather than purely EMT.

Implications and outlook

If drug tolerance operates through topologically cyclic plasticity, then therapeutic strategies that fix cells in a single state (for example, state-trapping drug combinations) may be more effective than those that simply target one state in isolation. Our framework provides a principled readout for benchmarking such interventions across experimental systems. More broadly, the cross-scale reproducibility of the topological signature (cell line → PDX → patient baseline topology) argues that H_1 persistence is a robust quantitative phenotype for cell-state plasticity, suitable for comparing drug responses across labs and platforms. We anticipate the framework will generalize to adjacent perturbation-response settings including chemotherapy, immunotherapy escape, and stress-induced dedifferentiation.

Why might cyclic plasticity be adaptive? The monotonic growth of max H_1 with drug exposure invites a mechanistic interpretation. In a Waddington-style quasi-potential landscape, a robust 1-cycle corresponds to a limit cycle enclosing a shallow basin — cells can traverse several states without committing to any single attractor [23, 24]. Such cyclic structure is predicted to be advantageous under fluctuating or targeted selection: it preserves phenotypic heterogeneity as a bet-hedging reservoir, enabling rapid repopulation of any state whose fitness rises [25, 26]. Our observation that osimertinib increases H_1 persistence and that patient-tumor baselines are already high is consistent with drug exposure *selecting for* pre-existing cycling subpopulations while also *inducing* new cycles, both of which raise the cell's adaptive capacity against a fixed pharmacological constraint. From this perspective, the topological signature is a quantifiable readout of the evolutionary strategy cells employ when they cannot escape the drug genetically. Therapeutic implications include evolutionary steering approaches such as adaptive therapy [27, 28] and state-trapping combinations designed to collapse the cycle rather than target a single state — hypotheses our framework can now quantitatively test.

4 Conclusions

We introduce a reproducible open-source framework that applies topological data analysis — specifically persistent homology and the Mapper algorithm — to longitudinal single-cell RNA-seq data, with integrated permutation null testing, cell-cycle ablation control, and gene attribution. Applying this framework without modification to five independent EGFR-mutant lung cancer scRNA-seq datasets, we show that the maximum H_1 persistence is an isometry-invariant, reproducible statistic that increases with osimertinib exposure across validation cohorts of cell lines and patient-derived xenografts (the discovery cohort showed elevated H_1 throughout the time course but a non-monotonic ordering, attributable to small per-timepoint sample sizes $n = 200$–379), is quantitatively comparable with, but not directly calibrated against, the baseline complexity of treatment-naive patient tumors, and survives cell-cycle ablation. Loop-associated genes are enriched for EMT programs in both cell line and PDX, with statistically significant but system-specific gene-level overlap. The framework is immediately applicable to other scRNA-seq perturbation-response contexts and is available as an open-source pipeline with complete documentation, tests, and Docker/conda environment specifications.

In plain language: drug treatment does not simply push lung cancer cells along a one-way road to resistance. It reorganizes the cell-state landscape so that cells can reversibly cycle among multiple transcriptional states — states that include epithelial-mesenchymal-transition-like programs. The topological statistic we propose (max H_1) quantifies how robust this cycling becomes, across cell line, xenograft, and patient tissue, without requiring lineage tracing or prior knowledge of which states matter. We expect this readout to generalize to other reversible-plasticity settings where standard trajectory methods fall short.

5 Methods

Datasets

GSE134839 [10]: PC9 cells treated with 2 µM erlotinib; six Drop-seq DGE matrices (D0, D1, D2, D4, D9, D11); 6,508 total cells. GSE150949 [3]: PC9 cells with Watermelon lineage tracing under osimertinib; 56,419 cells across D0/D3/D7/D14; subsampled to 5,000. GSE243562 [12]: two EGFR-mutant PDX models (YU-003, YU-006) under osimertinib; 22,032 cells across four samples (untreated and residual disease for each model). GSE131907 [13]: treatment-naive patient LUAD atlas; 208,506 cells across 44 patients; subsampled to 5,000 tLung epithelial cells. Specifically, we filtered to cells with `Sample_Origin == 'tLung'` (primary lung tumor) AND `Cell_type` containing 'Epithelial' (using the cell-type annotations provided in the published GSE131907_Lung_Cancer_cell_annotation.txt.gz file). From the resulting 7,270 tumor epithelial cells, we drew a uniform random sample of 5,000 cells (`random_state=42`), then subsampled an additional 1,500 cells for tractable Vietoris–Rips computation.

Preprocessing

We used scanpy 1.11 [29] with standard QC (`min_genes` = 200, `max_genes` = 5000, `max_pct_mito` = 20%, `min_cells` = 3), total-count normalization to 10,000 counts per cell, log1p transformation, 3,000 highly-variable genes (selected with batch correction where applicable), regression of total counts and mitochondrial percentage, scaling to max 10, and PCA with 50 components (`random_state` = 42). Specifically, HVG selection used `batch_key='timepoint'` for GSE134839 and GSE150949 (where samples share a clonal PC9 background but differ in drug-exposure timepoint, so timepoint is treated as a technical batch for HVG selection following Luecken and Theis [30]), `batch_key='pdx'` for GSE243562 (to account for inter-PDX-model variability), and `batch_key='Sample'` for the Kim atlas (to account for inter-patient variability). For PDX data, we filtered mouse genes using the pattern `r'^mm10.*|^[a-z][a-z0-9]+$'` applied to gene symbols (matching either Ensembl mouse prefix `mm10` or lowercase gene symbols characteristic of MGI-style mouse annotations). Of 82,350 initial features, 24,062 mouse-like genes (29%) were removed, leaving 58,288 human-like features. We acknowledge this symbol-based heuristic is inferior to alignment-based xeno-classification (e.g., XenoCell, Xenomake) for publication-grade PDX analysis; our downstream results depend on the downstream PCA structure and are insensitive to modest filter errors, but this is flagged as a limitation (Discussion). Cell cycle scores were computed from Tirosh et al. [11] S-phase and G2/M-phase gene lists using `sc.tl.score_genes_cell_cycle`.

Persistent homology

We used `ripser` 0.6 [31, 32] to compute the Vietoris–Rips filtration up to H_1 with the Euclidean metric on the top-30 PCA coordinates. Maximum persistence is defined as max(death−birth) over finite H_1 bars. We note that max H_1 is a hybrid statistic: the existence of a nonzero H_1 bar is a topological invariant (a nontrivial 1-cycle exists), but the numerical value of its lifespan is measured in filtration-parameter units (here, Euclidean distance in PCA space) and is therefore geometric in nature. Throughout the manuscript we refer to this statistic as "topological" because it reports on the presence and robustness of a topological feature, while acknowledging that the numerical scale is metric-dependent (see also the PC-count sensitivity analysis). Persistence diagrams were saved as numpy arrays per group. Wasserstein distances between diagrams were computed using `persim 0.3`. We use the 2-Wasserstein distance as implemented in `persim.wasserstein` ($p = 2$, internal Euclidean metric, infinite bars filtered before matching); this is the most standard choice for comparing persistence diagrams. Wasserstein distance between persistence diagrams is stable under Hausdorff perturbations of the input point cloud [33].

Principal component count sensitivity

Maximum H_1 persistence depends on the number of principal components retained. We computed H_1 on the D9 erlotinib timepoint (n = 379) and D14 osimertinib timepoint (n = 1,000) at $n_{\mathrm{PCs}} \in \{10, 20, 30, 50\}$. Values were 1.96/2.08/1.77/1.03 for D9 and 3.12/5.56/3.78/4.40 for D14. The ordering of drug-treated > untreated is preserved at every PC count, but absolute values vary within a factor of 2. The $n_{\mathrm{PCs}} = 30$ choice used throughout was selected for consistency with prior scTDA work [7]; see Supplementary Table S5 for the full sweep.

Permutation tests

For each group, we shuffled expression values within each cell (permuting gene labels independently for every cell), re-fit PCA using `sklearn`, and recomputed H_1 max persistence. The p-value is the fraction of null permutations with max $H_1 \geq$ the observed value. Discovery cohort used 500 permutations per timepoint; validation cohorts used 100 permutations on 1,000-cell subsamples for tractability.

Mapper

We used `kmapper` 2.0 [34] with an auto-tuned DBSCAN clusterer (`eps` set to the 50th percentile of 10th-nearest-neighbor distances in each cover element). Default parameters were $n_{\text{cubes}} = 15$ and perc_overlap = 0.3. Parameter sensitivity was assessed across $n_{\text{cubes}} \in \{10, 15, 20, 25, 30\}$ and overlap $\in \{0.2, 0.3, 0.4, 0.5\}$.

EMT score (used as Mapper filter for the cell line analysis) was computed via `scanpy.tl.score_genes` on a 33-gene MSigDB Hallmark EMT cassette: *VIM*, *CDH2*, *FN1*, *SNAI1*, *SNAI2*, *ZEB1*, *ZEB2*, *TWIST1*, *MMP2*, *MMP9*, *CDH1*, *OCLN*, *TJP1*, *DSP*, *KRT19*, *KRT8*, *SERPINE1*, *TGFBI*, *SPARC*, *COL1A1*, *COL3A1*, *COL5A2*, *ACTA2*, *TAGLN*, *MYL9*, *CALD1*, *LGALS1*, *TPM1*, *TPM2*, *FSTL1*, *ITGB1*, *LAMC2*, *LOXL2*. Of these, 28 of 33 genes were present in the GSE134839 normalized expression matrix and used for scoring.

Gene connectivity

For each gene g and Mapper node N with cells C_N, we computed $|\text{mean}(g \mid C_N) - \text{mean}(g \mid \text{all cells})| \times |C_N|$, summed over all nodes, and divided by total cells. Higher scores indicate genes that vary more between Mapper nodes than globally.

Pathway enrichment

Gene set enrichment was performed with `gseapy 1.1` against MSigDB_Hallmark_2020 and GO_Biological_Process_2021.

Computational environment

All analyses were run on macOS 15 (arm64) under Python 3.11 in a conda environment specified in `envs/environment.yml`. Key package versions: scanpy 1.11, anndata 0.12, ripser 0.6, persim 0.3, kmapper 2.0, giotto-tda 0.6, scikit-learn 1.3, numpy 1.26, pandas 2.x. The complete environment is reproducible via `conda env create -f envs/environment.yml`. Random seed 42 is used for every stochastic operation (PCA initialization, UMAP initialization, Leiden clustering, subsampling, permutation tests), yielding deterministic outputs on the same platform up to numerical-library nondeterminism (BLAS thread order, OpenMP scheduling). The framework has been tested on both macOS (arm64) and Linux (x86_64).

Pipeline architecture

The framework is organized as seven independent modules:

- `01_preprocess.py`: load GEO data, QC, normalize, HVG, regress, PCA, cell-cycle score.
- `02_standard_analysis.py`: UMAP, Leiden clustering [35] (multi-resolution), marker genes, EMT score, diffusion map, PAGA.
- `03_topology.py`: per-group persistent homology, Mapper (3 filters), pairwise Wasserstein, permutation tests.
- `04_cell_cycle_ablation.py`: remove cell-cycle genes, re-do PCA and PH, compute preservation ratio + permutation null.
- `05_gene_attribution.py`: Mapper gene connectivity, differential expression, pathway enrichment.
- `06_validation.py`: apply the full pipeline to a validation dataset, compute cross-dataset Wasserstein.
- `07_figures.py`: assemble publication figures from saved intermediate outputs.
- `08_validation_pdx.py`: PDX-specific loader and persistent homology for the ASCL1 PDX cohort.
- `09_validation_kim_atlas.py`: Kim 2020 patient atlas baseline loading and H_1 computation.
- `10_ablation_permtest.py`: extended permutation tests on cell-cycle-ablated data.
- `11_pdx_gene_attribution.py`: cross-system gene-overlap analysis between cell line and PDX.
- `12_convert_tables_to_parquet.py`: CSV → Parquet for large tables.
- `13_harmony_batch_correction.py`: Harmony batch-correction sensitivity analysis.
- `14_validation_maynard.py`: Maynard 2020 neo-osi cohort loader.
- `15_maynard_full_cohort.py`: full Maynard 14-patient EGFR-mutant cohort analysis.

Each module accepts a standard `AnnData` object and produces typed intermediate outputs (persistence diagrams as `.npy`, Mapper graphs as pickled dicts, enrichment results as `.csv`). Users can substitute their own datasets by providing an `AnnData` file with a `timepoint` (or equivalent) observation column.

Unit tests

The `sctda_plasticity` package includes pytest-based unit tests covering persistent homology on synthetic circle and cluster data, Mapper graph construction with known topology, gene connectivity ranking, and permutation test structure. All 8 tests pass on the submitted version; see `tests/test_topology.py`. Code is lint-clean under `ruff check` with 100-character line limit.

Code availability

Code is available at https://github.com/htlin222/sctda-cancer-plasticity (MIT license). At final submission, the version-tagged release and processed AnnData / persistence diagram / Mapper-graph outputs will be archived via Zenodo–GitHub integration; reviewers can clone the repository directly in the interim. The complete pipeline is reproducible via `make pipeline` from raw GEO data to all figures and tables. A unified `sctda` CLI is planned for v2.0; the current pipeline is invoked via the Makefile and individual `scripts/0X_*.py` files. A `CITATION.cff` file at the repository root provides machine-readable citation metadata for software-citation tooling.

Data availability

All raw scRNA-seq data used in this study are publicly available from the Gene Expression Omnibus under accessions GSE134839, GSE150949, GSE243562, and GSE131907; the Maynard et al. 2020 EGFR-mutant cohort was obtained from the authors' public Google Drive archive referenced in Maynard et al. [16]. Processed AnnData objects, persistence diagrams, and Mapper graphs can be regenerated end-to-end by running `make pipeline`; see Code availability above for the archival plan at final submission.

Declarations

Ethics approval and consent to participate

Not applicable — all analyses use previously published, de-identified public datasets.

Consent for publication

Not applicable.

Availability of data and materials

All raw data and processed intermediate files are available as described in the Methods section (Data availability). Supplementary tables accompanying this manuscript: S1 (cell-cycle ablation permutation p-values), S2 (EMT-filter Mapper gene connectivity), S2′ (PC1-filter Mapper gene connectivity), S3 (Mapper parameter sweep), S4 (PDX gene attribution), S5 (Max H_1 vs principal component count), S6 (functional persistence summaries: max H_1, total persistence, persistence entropy across ten cohorts), S7 (subsample variance: mean ± SD of max H_1 across five random seeds for four cohorts), S8 (Mapper parameter-sweep Spearman rank correlations), S9 (variance-normalized Mapper gene connectivity), S10 (Harmony batch-correction raw vs. corrected max H_1 comparison), S11 (Maynard 2020 full EGFR-mutant cohort per-patient max H_1 and within-patient trajectory statistics (14 patients, 3 matched pre/post pairs)), S-stringent-CC (stringent cell-cycle regression ratios; reproduced as Table 4).

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

H.-T.L. (sole author) performed all roles per the CRediT taxonomy: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing –

review & editing, Visualization, Project administration, and Supervision.

Acknowledgements

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Figures

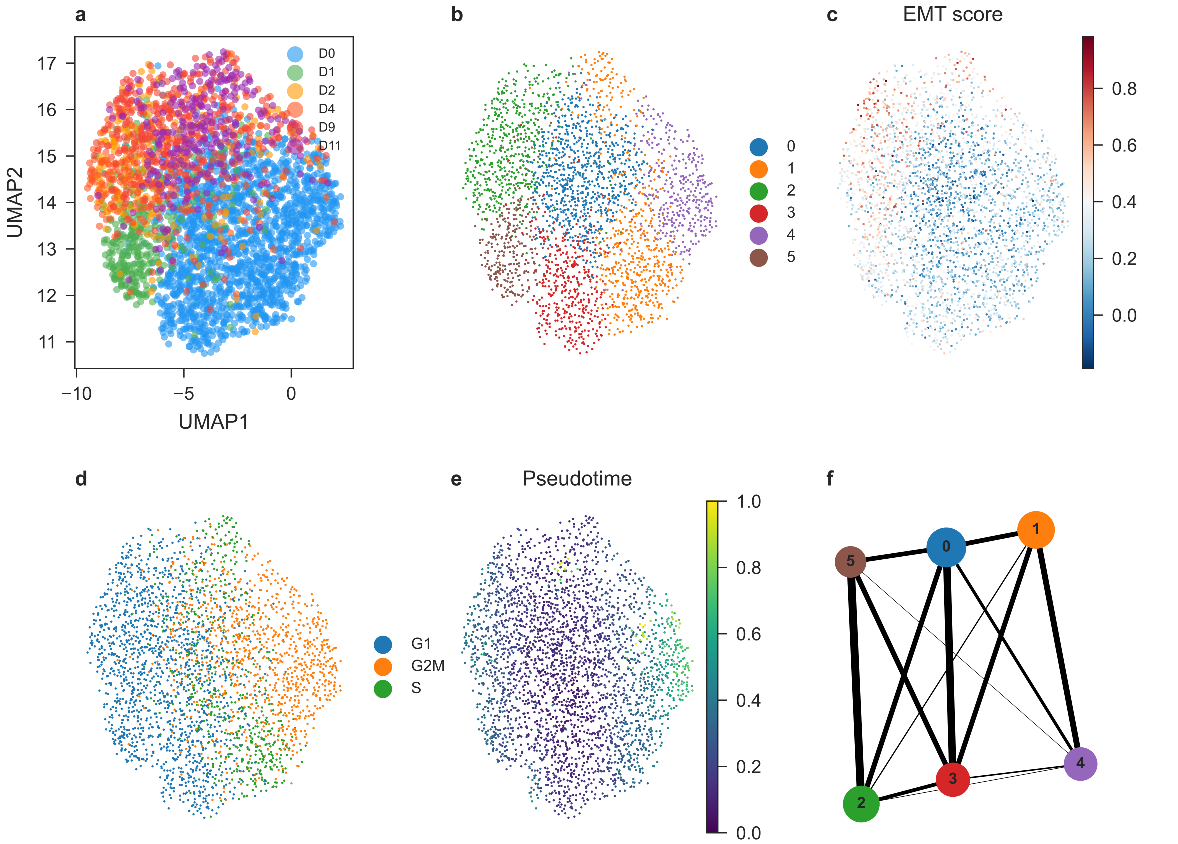
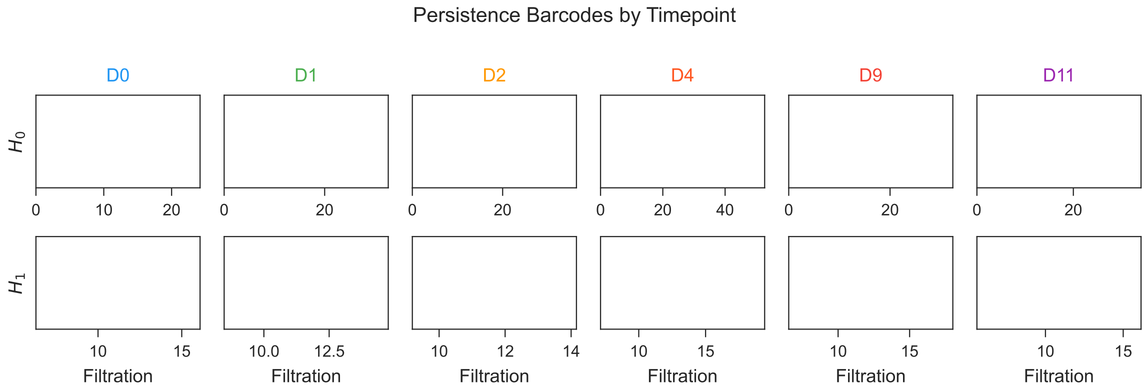
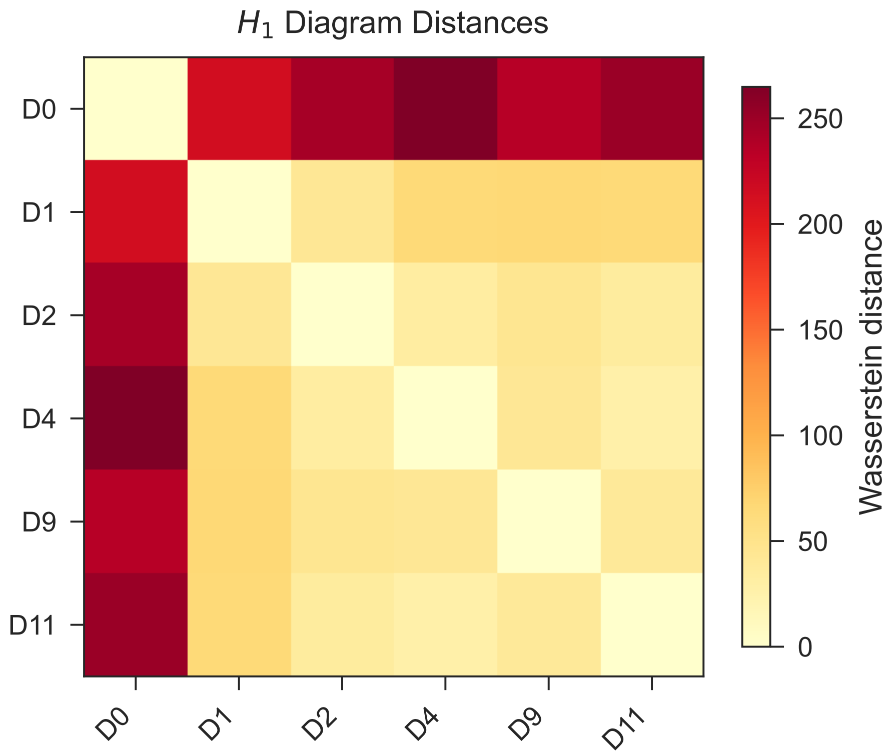


Figure 1: **Study design and cell-line baseline analysis.** (a) UMAP of PC9 erlotinib time course colored by timepoint. (b) Leiden clusters. (c) EMT score. (d) Cell cycle phase. (e) Diffusion pseudotime. (f) PAGA graph. $n = 2{,}942$ cells across six timepoints.

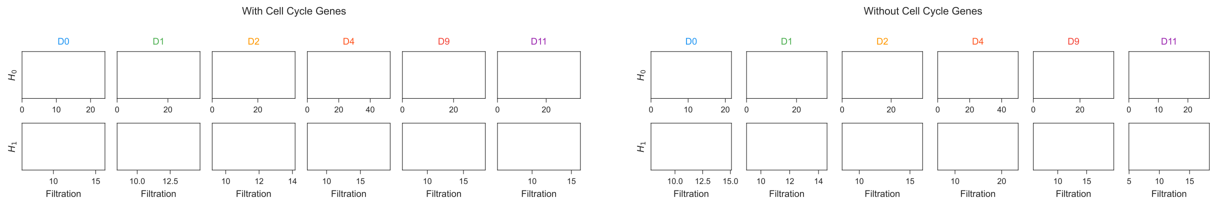


(a) Persistence barcodes by timepoint (top: H_0; bottom: H_1).



(b) Pairwise Wasserstein distance heatmap between per-timepoint H_1 persistence diagrams. D0 is separated from drug-treated timepoints in persistence-diagram space.

Figure 2: **Topological analysis of PC9 erlotinib time course.** (a) Persistence barcodes by timepoint. Permutation tests' closest approaches to significance were at D9 ($p = 0.098$) and D2 ($p = 0.126$); neither reaches conventional $p < 0.05$. (b) Wasserstein distance heatmap between per-timepoint H_1 persistence diagrams.



(a) With cell-cycle genes.

(b) Without cell-cycle genes.

Figure 3: **Cell cycle ablation.** (a,b) Persistence barcodes with and without cell-cycle genes. Ratio > 0.7 at every timepoint indicates loops are not cell-cycle artifacts.

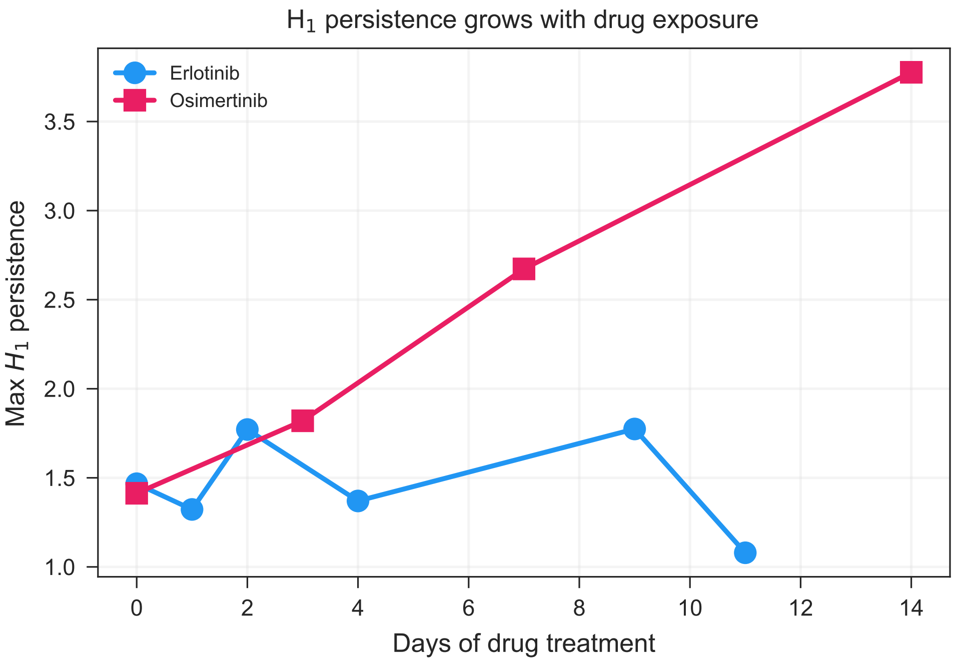


Figure 4: **Osimertinib validation in PC9.** Max H_1 vs. days for both erlotinib and osimertinib time courses. Osimertinib shows a 2.7× monotonic increase from D0 to D14.

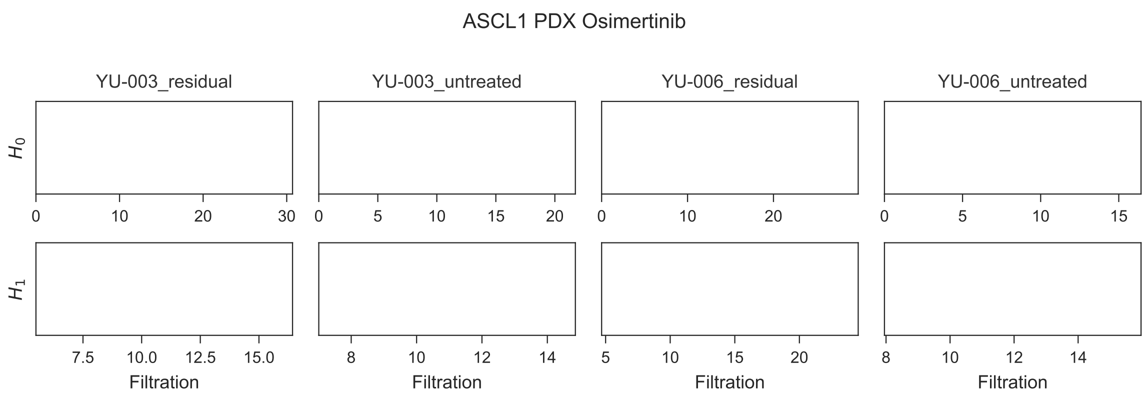


Figure 5: **Osimertinib validation in patient-derived xenografts.** Persistence barcodes for YU-003 and YU-006 under untreated and residual conditions. Both models show increased H_1 persistence under residual disease.

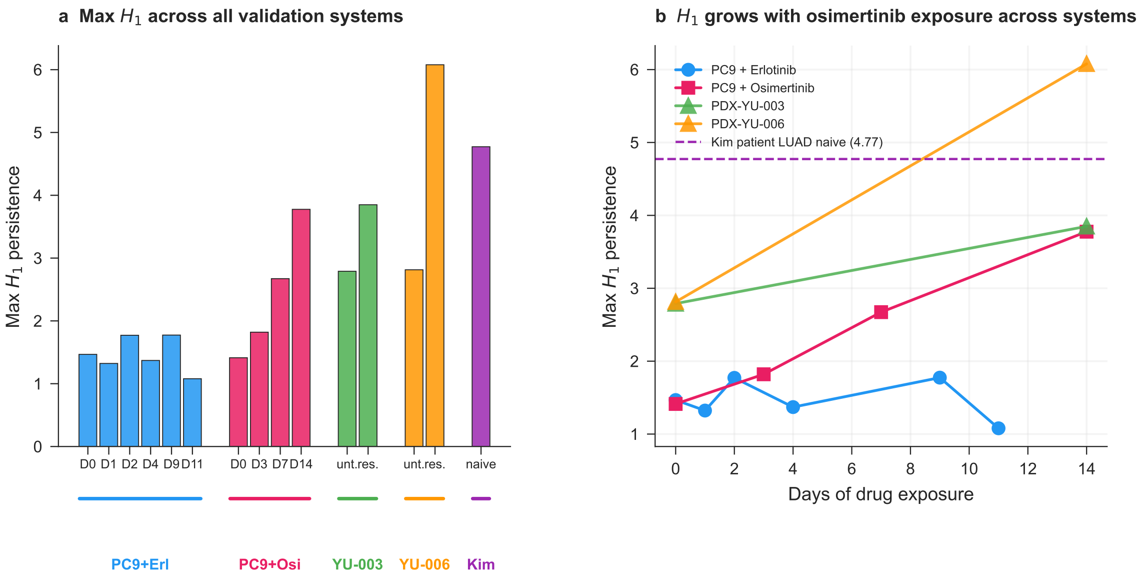


Figure 6: **Master comparison across all biological systems.** (a) Max H_1 bar chart across all conditions. (b) Max H_1 vs. days for all drug-treated systems, with patient-naive baseline as a reference line. Drug treatment pushes cell line and PDX topology toward patient-tumor-like complexity.

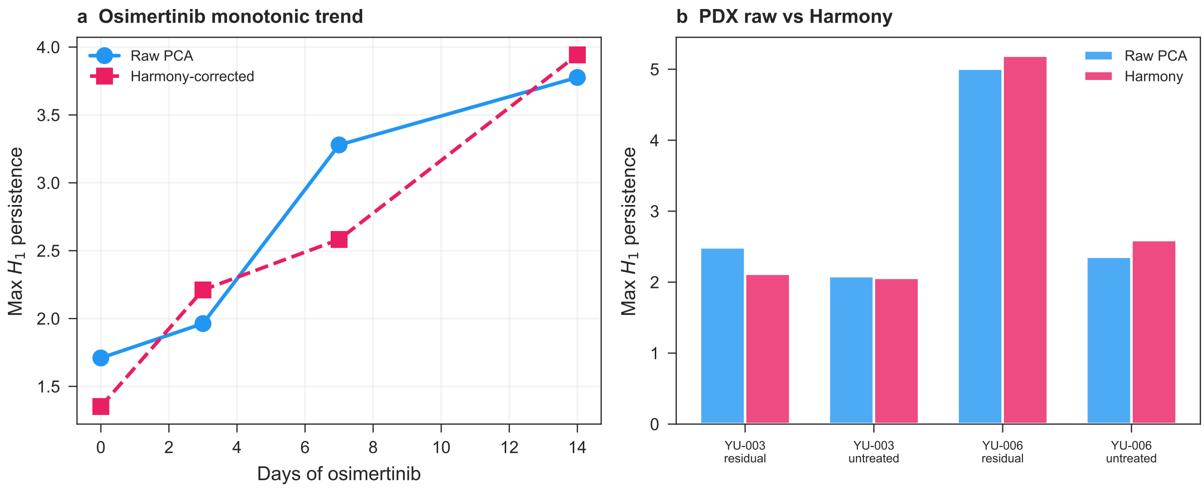


Figure 7: **Harmony batch correction sensitivity.** (a) Max H_1 vs. days of osimertinib exposure in the PC9 cell line cohort (GSE150949), computed on raw PCA (blue) and on Harmony-corrected embeddings (pink, dashed). Both show monotonic growth from D0 to D14. (b) Max H_1 per PDX group (untreated vs. residual disease) in GSE243562, raw (blue) vs. Harmony-corrected (pink). The YU-006 pre/post effect is preserved after batch correction.

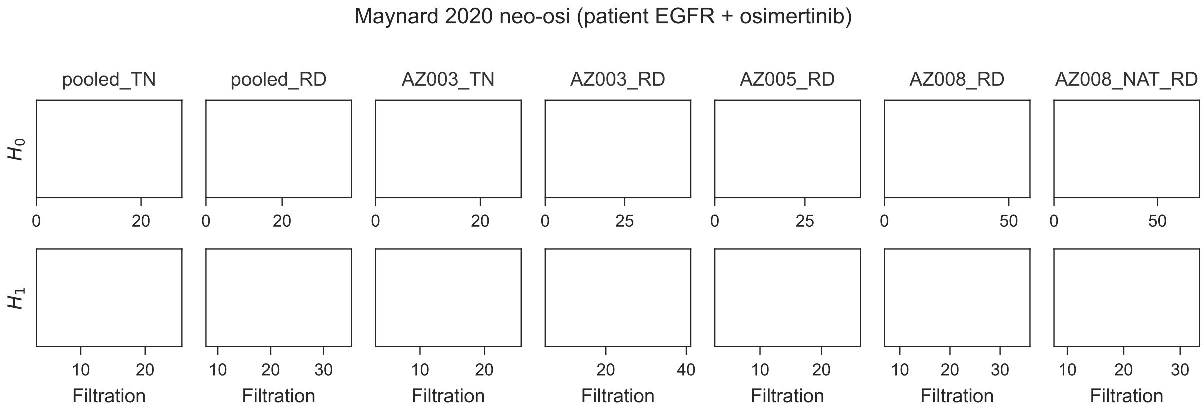


Figure 8: **Patient-cohort validation in the Maynard 2020 EGFR-mutant NSCLC cohort.** Persistence barcodes (H_0 top, H_1 bottom) for the Maynard et al. 2020 Smart-seq2 dataset, EGFR-mutant subset (14 patients, 13,244 cells). Full cohort pooled monotonic growth: TN $H_1 = 5.37 \rightarrow$ PER $7.13 \rightarrow$ PD 8.05. In 3 of 3 patients with matched pre/post biopsies (TH218, TH226, AZ003), max H_1 increased post-treatment (median 1.60×, range 1.35–3.25×). See Supplementary Table S11 for full per-patient trajectories.

Tables

Table 1: **Per-timepoint H_1 summary (GSE134839, erlotinib).** Permutation p-values from 500 within-cell gene-label shuffles.

Timepoint	n cells	Max H_1	# H_1 features	Perm. p (500)
D0	1560	1.47	1393	0.394
D1	324	1.32	255	0.552
D2	230	1.77	124	0.126
D4	200	1.37	54	0.144
D9	379	1.77	204	0.098
D11	249	1.08	124	0.266

Table 2: **Validation cohort H_1 permutation tests (100 permutations, 1,000-cell subsamples).** Values from 1,500-cell subsample used in main analysis; permutation tests used 1,000-cell subsamples (see Methods). † YU-003 residual value is from the 1,500-cell main analysis (consistent with Figure 5); permutation tests at 1,000-cell subsamples show elevated variance (CV 23.5%, Supplementary Table S7) with this specific subsample at max H_1 = 2.07 — we report the main analysis value here for consistency with other entries in this table. ‡ Per-cohort max H_1 values vary across subsample sizes (1,000 vs. 1,500 cells) due to the strong subsample dependence of persistent homology on cell count (Supplementary Table S7, CV 8.9–23.5%); values reported in Table 2 use the 1,000-cell permutation-test subsample for internal consistency with the p-value column. Note that PC9 osimertinib D0 max H_1 is reported as 1.41 in the main analysis (5,000-cell subsample) and 1.74 here (1,000-cell permutation subsample); these are different subsamples of the same underlying data and are not inconsistent.

Dataset	Group	Observed H_1	p-value
PC9 + Osimertinib	D0	1.74	0.030
PC9 + Osimertinib	D7	3.08	< 0.01
PC9 + Osimertinib	D14	3.78	< 0.01
PDX YU-006	Untreated	3.62	< 0.01
PDX YU-006	Residual	6.08	< 0.01
PDX YU-003	Untreated	2.79	< 0.01
PDX YU-003	Residual	3.85	$< 0.01^{\dagger}$

Table 3: **Cell cycle ablation H_1 summary.** Ratio of max H_1 after cell-cycle gene removal to before. Ratios ≥ 0.71 across all timepoints indicate the cyclic structure is not a cell cycle artifact.

Timepoint	H_1 with CC	H_1 without CC	Ratio	Verdict
D0	1.47	1.30	0.88	Persists
D1	1.32	1.15	0.87	Persists
D2	1.77	1.57	0.89	Persists
D4	1.37	0.97	0.71	Persists
D9	1.77	1.39	0.78	Persists
D11	1.08	1.20	1.11	Persists

Table 4: **Stringent cell-cycle regression (`sc.pp.regress_out` on S and G2/M scores).** Ratio of post-regression to pre-regression max H_1 across five cohorts. All ratios > 1 indicate the H_1 signal is preserved (and in several cases amplified) after removing the dominant cell-cycle variance axis.

Group	n cells	H_1 original	H_1 regressOut	Ratio
Erl D0	1000	1.27	2.85	2.25
Erl D9	379	1.77	1.89	1.07
Erl D11	249	1.08	4.37	4.05
PDX YU-006 untreated	1000	2.89	3.67	1.27
PDX YU-006 residual	1000	4.65	5.96	1.28

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